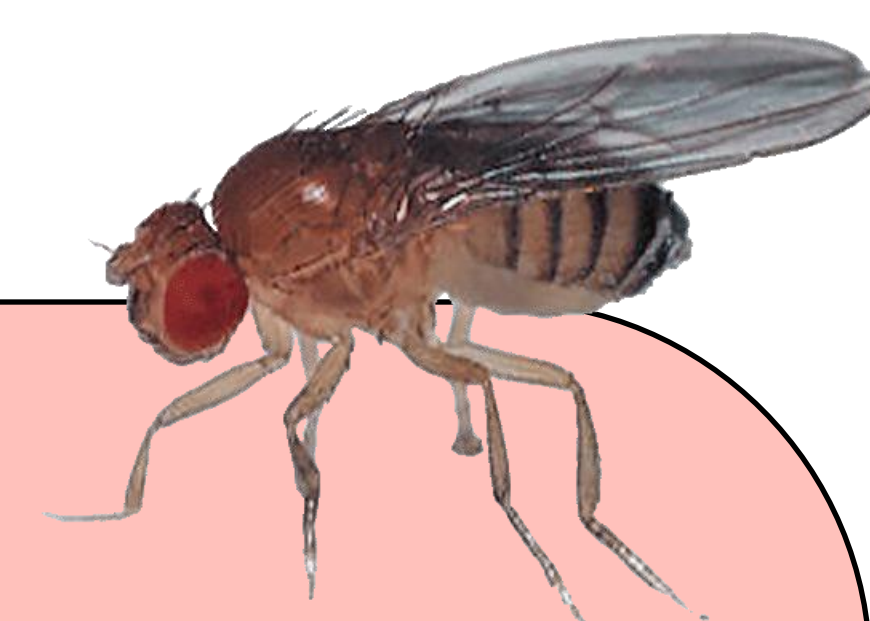


Investigating the Effects of Aspartame on *Drosophila* Physiology and Lifespan

Jordan Hoke, Maddy Liljenberg, JD Shaposchnick, James Kezos

Funded by Hologic



Background

Cardiovascular disease is currently the leading cause of death in the United States, responsible for approximately 20% of deaths. Obesity is one of the most significant and notable risk factors for heart disease, as well as other diseases such as Type II Diabetes and stroke. As obesity rates continue to climb in the United States, many companies have replaced traditional sugars with alternative, low-calorie sweeteners like Aspartame. Aspartame is a non-nutritive sweetener approximately 180-200 times sweeter than normal sugar and is widely used in, but not limited to, diet products. While considered generally safe by regulatory agencies like the Food and Drug Administration, emerging research suggests that Aspartame as well as other artificial sweeteners have a potential link to neurological, metabolic, and cardiovascular health risks.

Introduction

- The use of artificial sweeteners, such as aspartame, has significantly increased in the past decade to combat obesity and obesity-associated diseases.
- The scientific community continues to question how chronic aspartame exposure impacts biological systems.
- Despite regulatory approval (e.g., FDA, EFSA), legal status and safety concerns around aspartame have fluctuated over time.
- Drosophila melanogaster* (fruit flies) was used as a model due to genetic and developmental similarities to humans
- Fruit flies have been widely used as a model system for measuring dietary and toxicity effects (e.g. high-fat diets, high sugar diets, urea, ammonia, lithium)
- Yeast acts as a stimulant in flies due to its high protein and carbohydrate content, increasing ATP production.
- Hypothesis:** Aspartame exposure will negatively affect motor function, heart function, and lifespan in *Drosophila*.

Methodology

Mortality Assay

- Wildtype fruit flies were randomly separated and assigned one of five dietary treatments (Control, 0.01 mM Asp, 0.1 mM Asp, 1 mM Asp, and 10 mM Asp)
- ~500 flies per treatment group housed in separate cages.
- Each treatment was replicated 3 times (total of 15 cages).
- Aspartame dissolved in nanopure water and mixed into standard yeast paste (yeast, acetic acid, water).
- Dead flies were removed, sexed, and counted on a daily basis. (Yeast dishes changed daily)
- Mortality tracked daily for 42 days per replicate

Motor Function Assay

- Per treatment, 3 vials of 20 males and 3 vials of 20 female were placed in marked vials to measure climbing ability
- Negative geotaxis (climbing) assay performed 3 times per vial.
- Percent of flies to reach milestones at each 1 second intervals were recorded (2-cm, 5-cm, and 10-cm mark)



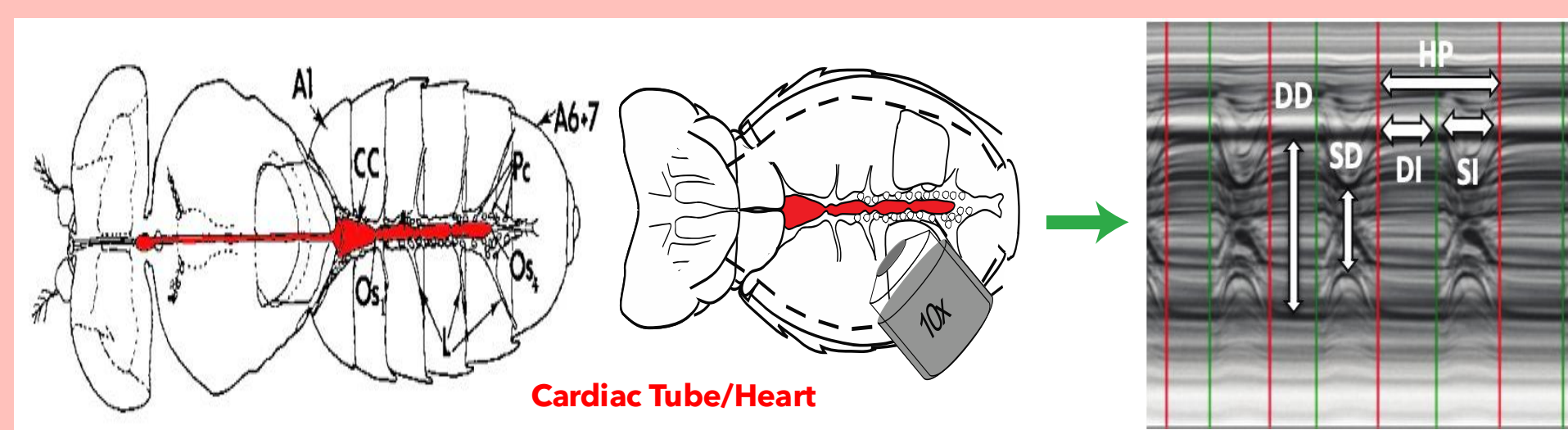
Feeding Preference Assay

- Additional cages of 500 flies were created and given two different Asp treatments that were stained with red or blue food dye
- After 3 days of consumption, 100 males and 100 females were randomly selected, separated by abdomen color and counted.



Cardiac Function Assay

- High-speed, fluorescent, video microscopy of fly hearts after 1, 3 and 5 weeks of exposure to dietary treatment (~20 flies per treatment were filmed and analyzed)
- Video processing and analysis completed via RStudio (generates 25 cardiac parameters)



Results

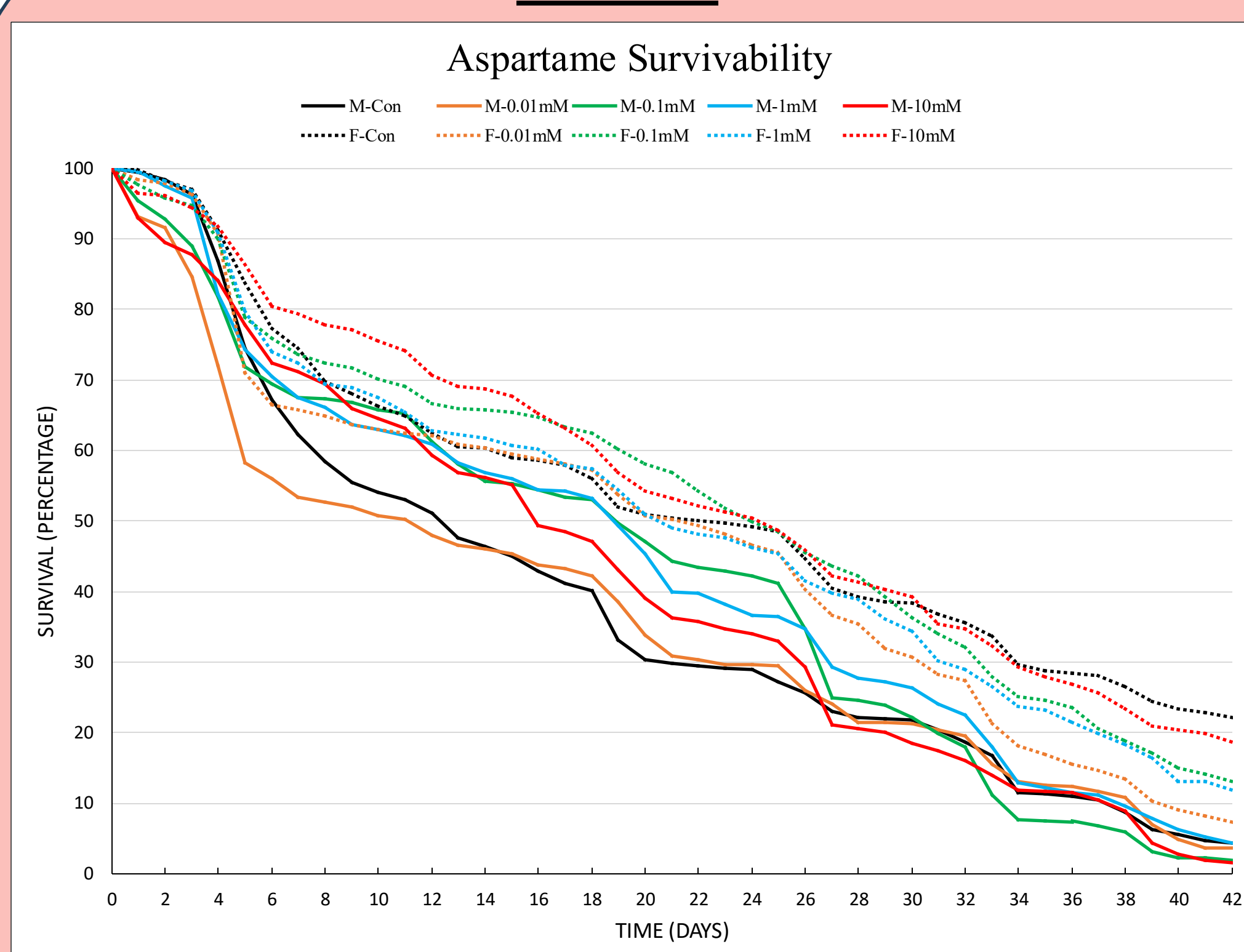


Figure 1. Aspartame survival curves of *Drosophila melanogaster* exposed to aspartame across 42 days, comparing males (solid lines) and females (dashed lines) across treatment groups.

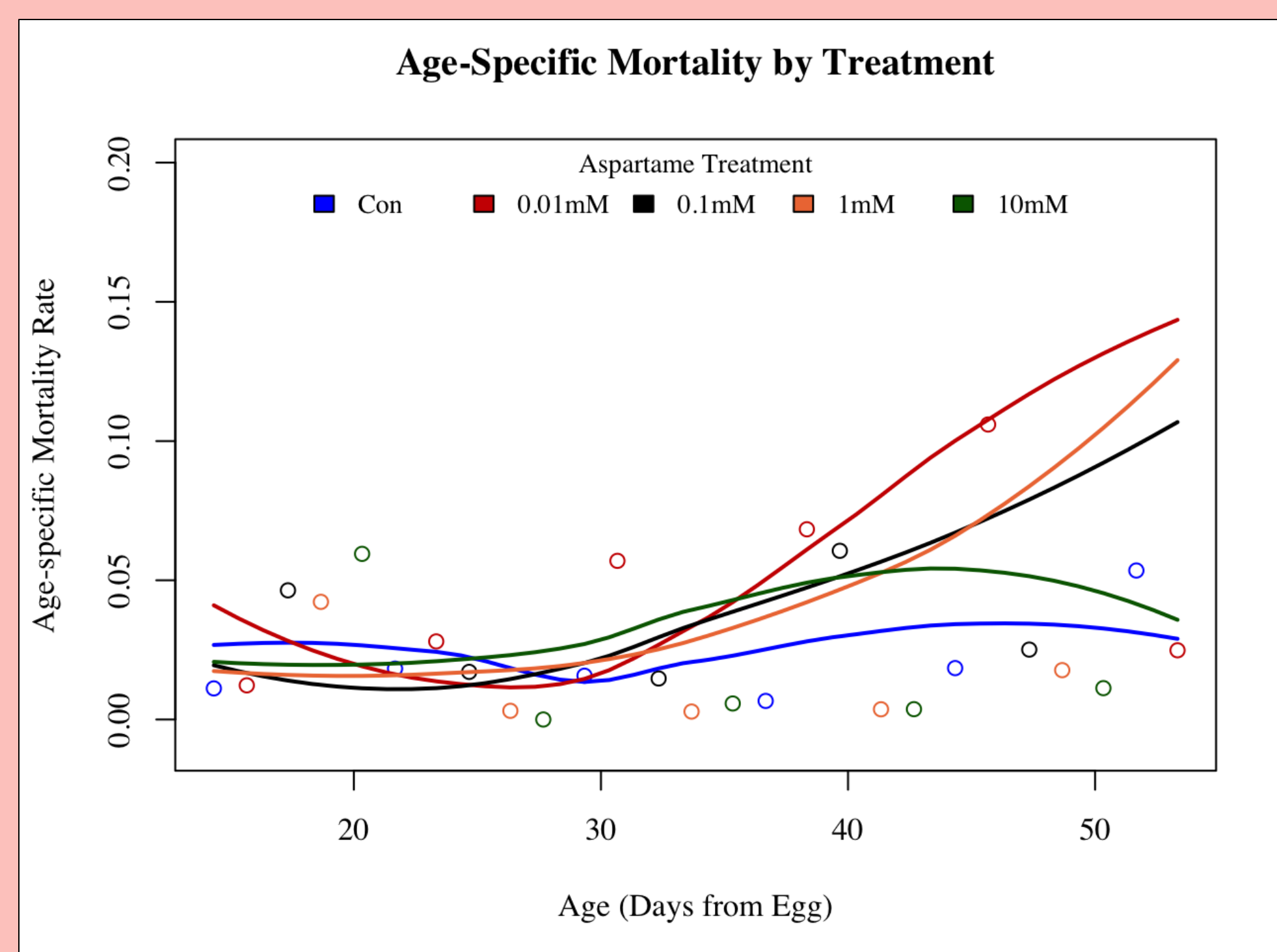


Figure 2. Age-specific mortality rates in female *Drosophila* across five treatment groups over 42 days. Each dot represents a 3-day average rate.

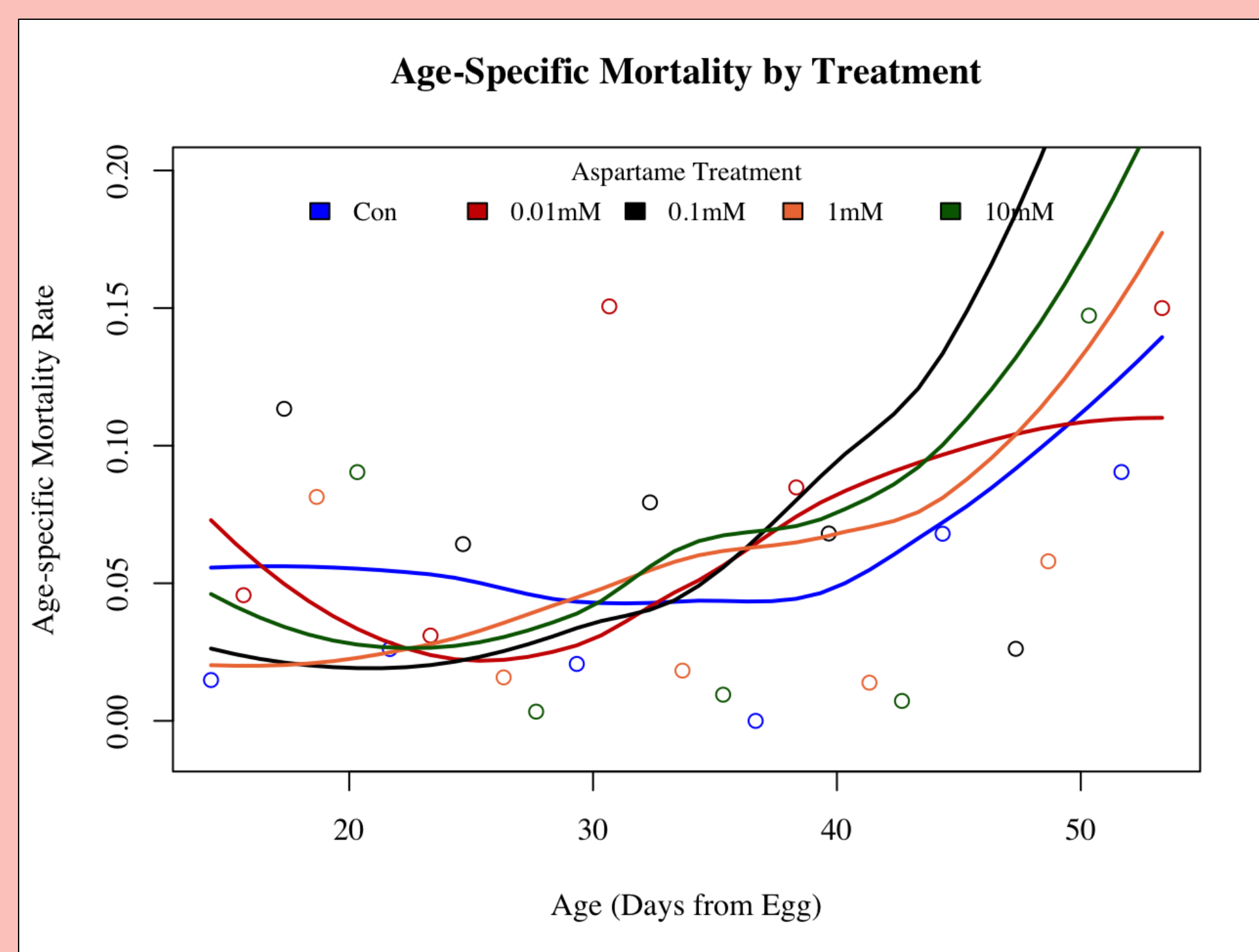


Figure 3. Age-specific mortality rates in male *Drosophila* across five treatment groups over 42 days. Each dot represents a 3-day average rate.

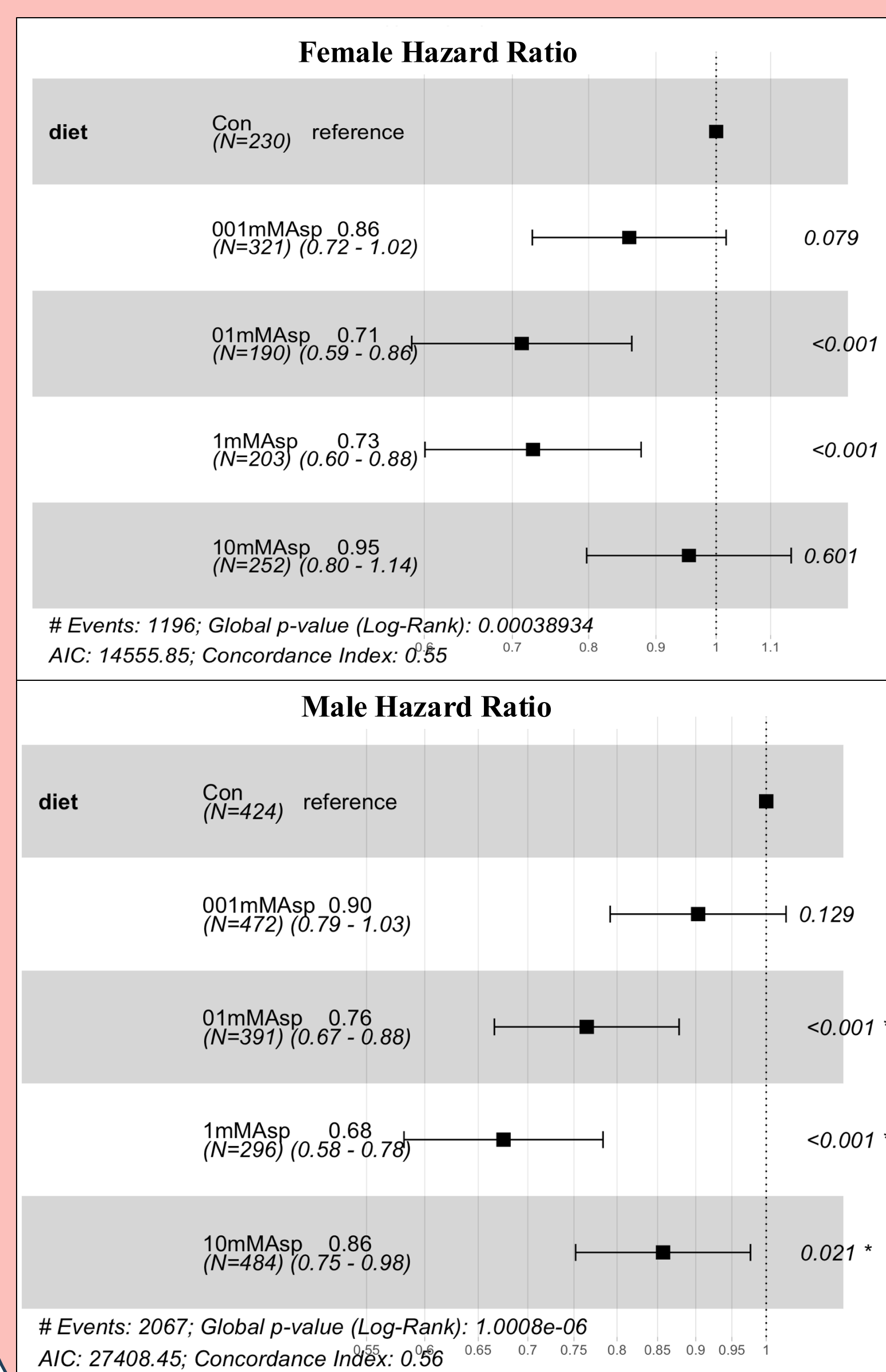


Figure 4. Cox-Hazard ratio Aspartame diets in female (top) and male (bottom) flies. Control diet in each sex is used as the reference (set to 1). A hazard ratio less than 1 equals less risk, whereas a hazard ratio greater than 1 equals more risk.

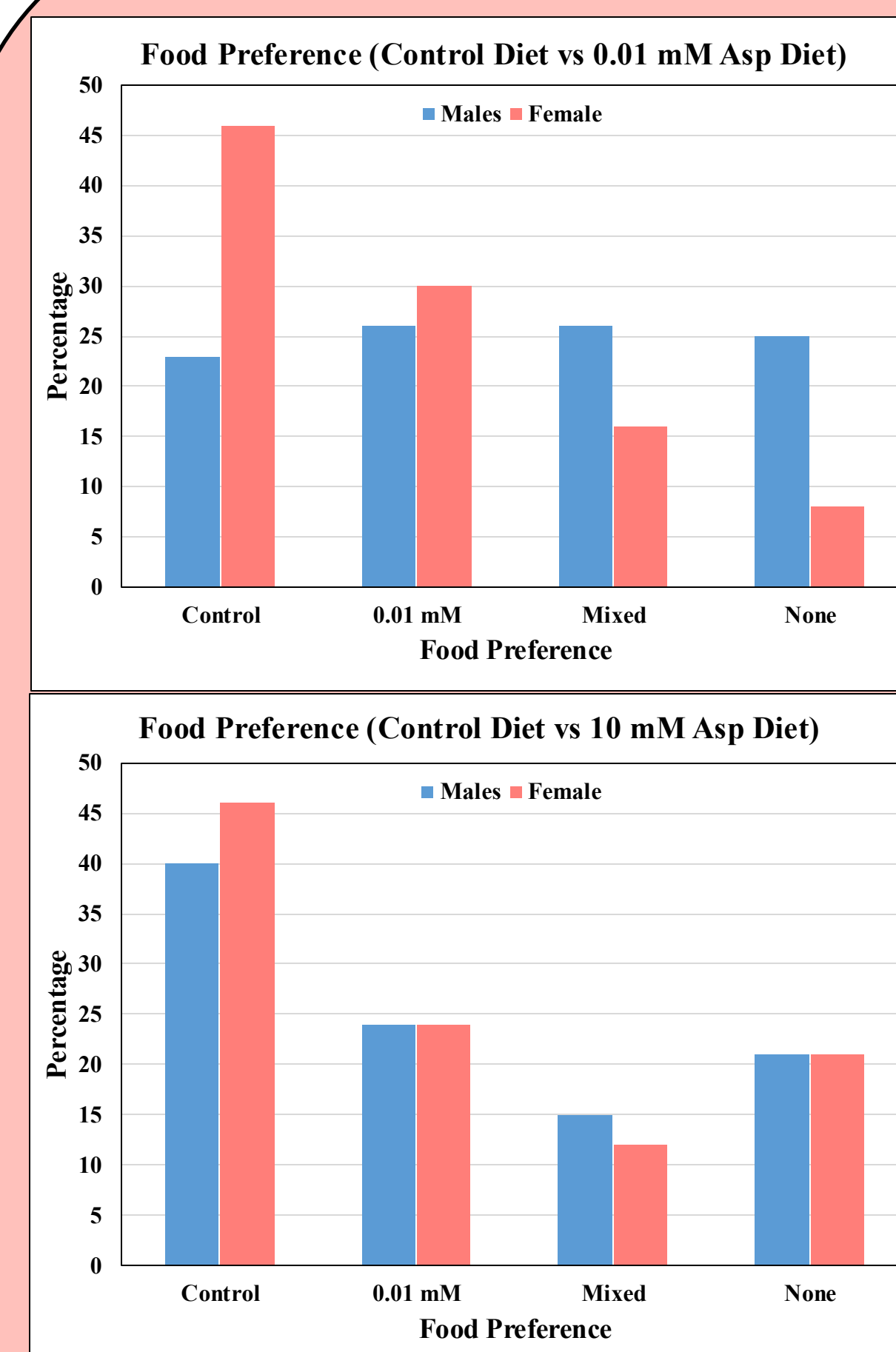


Figure 5. Feeding preference assay between two of the five Asp dietary treatments. In most comparisons, females and males preferred the diet with no Asp or lesser of the two Asp concentrations.

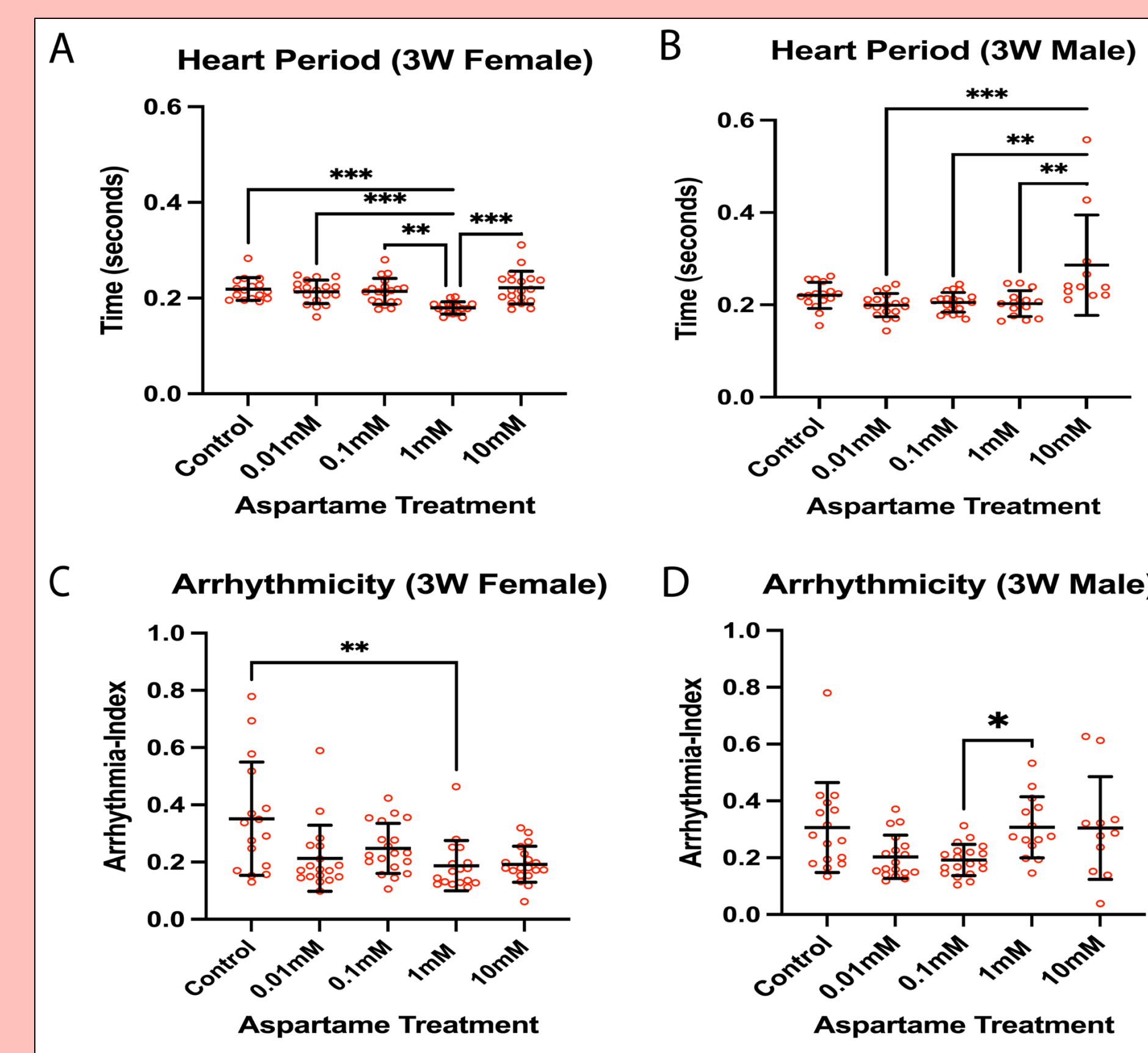


Figure 6. Effects of aspartame on cardiac function in 3-weeks old female and male flies. (A, B) Minimal effect on heart period in both sexes. Observed significant effects do not explain differences in mortality rates or lifespan. (C, D) Slight upward trend in arrhythmia in flies fed the control, non-Asp yeast diet (Mean $\pm$ SD plotted; Wilcoxon Rank-Sum Test, p-value < 0.05)

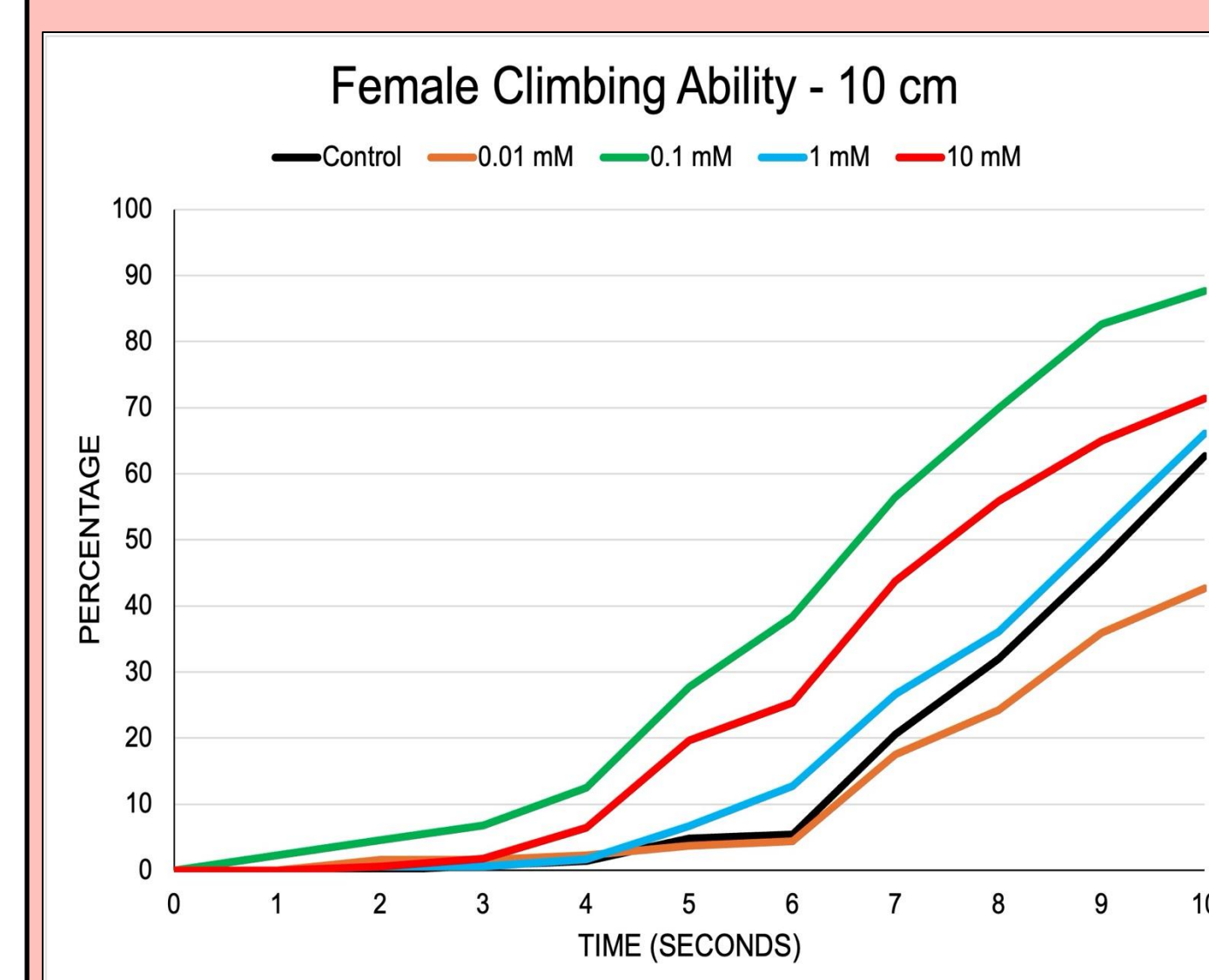


Figure 7. Percentage of female flies reaching 10 cm mark in climbing assay across time (seconds). No clear trend as you increase or decrease aspartame concentration in both female and male flies.

Conclusions

- Flies fed control diet had increased early-age mortality rates and an overall decrease in lifespan.
- 0.1 mM Asp and 1 mM Asp diets had lowest hazard ratio scores, implying potential lifespan benefits.
- Minimal effects on cardiac function or motor function due to Aspartame consumption (potentially reduces arrhythmicity).
- Aspartame triggers *Drosophila* bitter taste receptors, so possibly witnessing the *effects of caloric restriction* (flies preferred control yeast or lower Asp concentration diets).
- Previous studies have shown that 30-40% caloric restriction can increase lifespan and have other physiological benefits.

Future Studies

- Measure changes in overall lipid, glycogen, water, and protein content on select dietary treatments
- Investigate if aspartame consumption significantly alters transcriptome and metabolome profiles (Control vs 0.1 mM vs 10 mM diets)

Acknowledgments

We would like to acknowledge and thank **Hologic**, **CSTEM**, and **CSUSM** for the financial and research support for this project. We would also like to thank our fellow undergraduate research students who helped performed assays and collected data.